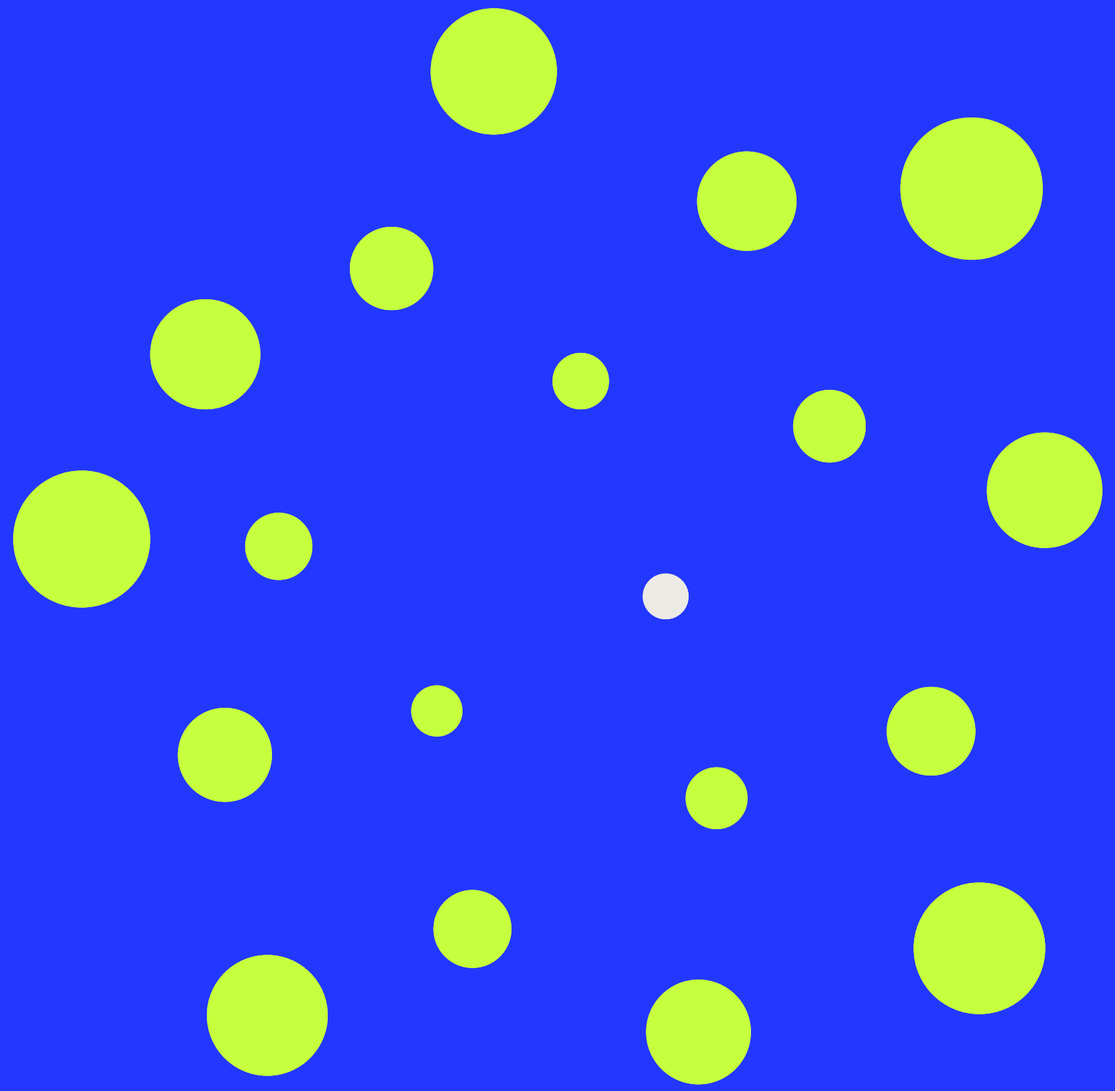
IDSN 599 · PRODUCT MANAGEMENT · TEAM 10

ecotein

Plastic In → Protein Out

Yasmin Aref · Paul Kang · Seraph Kumar · Adara Lui · Naomi Meininger





ECOTEIN PACKAGE DESIGN

ecotein

Protein made from dried microbes & plastic waste

The Process

01 **Extract**
Source plastic from recyclable materials

02 **Transform**
Break down plastic into building blocks for microbes

03 **Ferment**
Grow protein from microbes

04 **Dry into powder**
Shape into form



Unique Value Proposition

Ecotein provides nutritional needs at scale without relying on traditional agriculture, combining social responsibility with economic viability in the face of 2 global challenges

673M

people went hungry in 2024;
about 1 in every 12 people lack reliable access to food



FAO, SOFI 2025

50%

habitable land is already farmed; there's little room left to grow more food.



Our World in Data, via WEF 2019

Go-To-Market (GTM) Strategy

Ecotein succeeds with exponential technology and emerging market conditions.

THE TECHNOLOGY · EXPONENTIAL DRIVERS

Pyrolysis
Ideal for countries with limited agriculture

AI & Genomics
Microbes can be modified for higher yield and lower cost.

AI & Bioinformatics (vision)
Microbes could be tuned to fit each customer's nutritional needs

THE MARKET

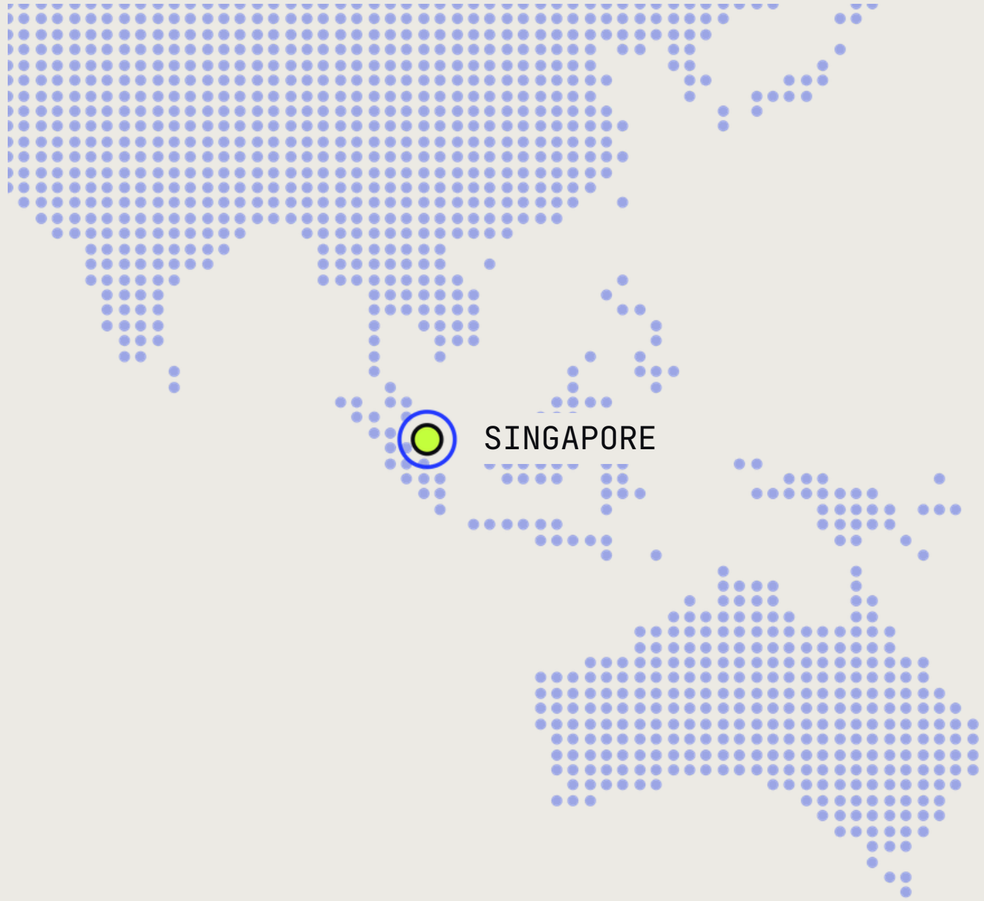
Alternative Protein Demand
Governments funding for alternative protein has grown from ~$700M in 2021 to $2.5B+ in 2025

Plastic Everywhere
Plastic is hard to recycle which creates a growing cost.

Plastic costs money to dump
S$89.60 a tonne to burn

Entry Point: Fish Feed in Singapore

Ecotein starts as an ingredient for fish and shrimp feed mills, in a market that pays most for protein.



The Pathway

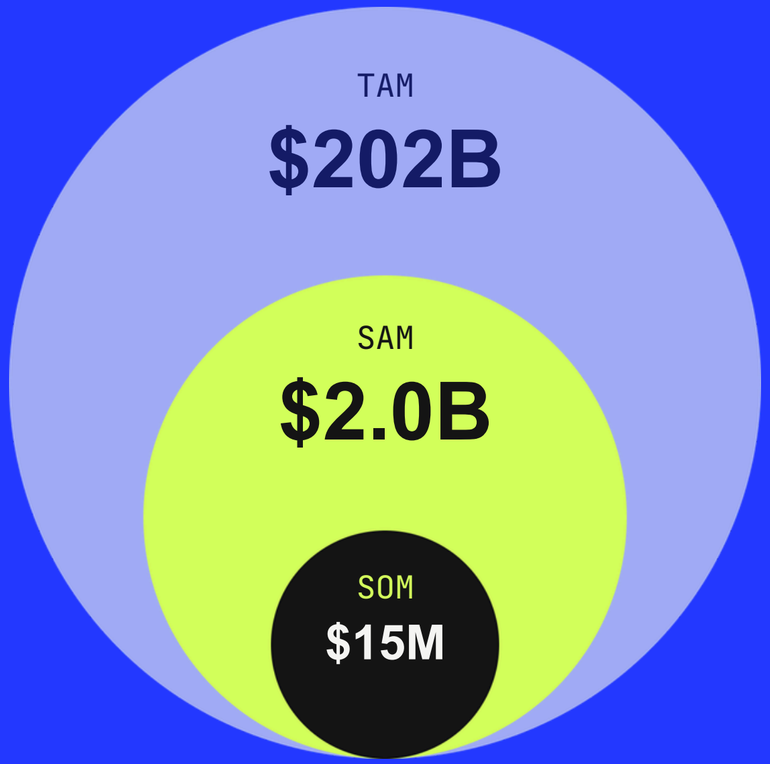
01 **Government funding and support**
Grants from Enterprise SG, A*STAR and Singapore Food Story

02 **Lower Barrier of Entry for animal feed**
Obtain feed licenses rather than novel-food approval

03 **Readily available plastic**
Source plastic from bottle returns at low or no cost

SINGAPORE → APAC → WORLD AT LARGE

Opportunity in the Protein Market



TAM $202B

All protein for human food ($55B) plus all the protein in fish feed and pet food ($147B), worldwide.

SAM $2.0B

Asia-Pacific fishmeal we can replace ($1.9B), plus aid food we can test into with WFP and UNICEF ($0.14B). 1.0% of TAM.

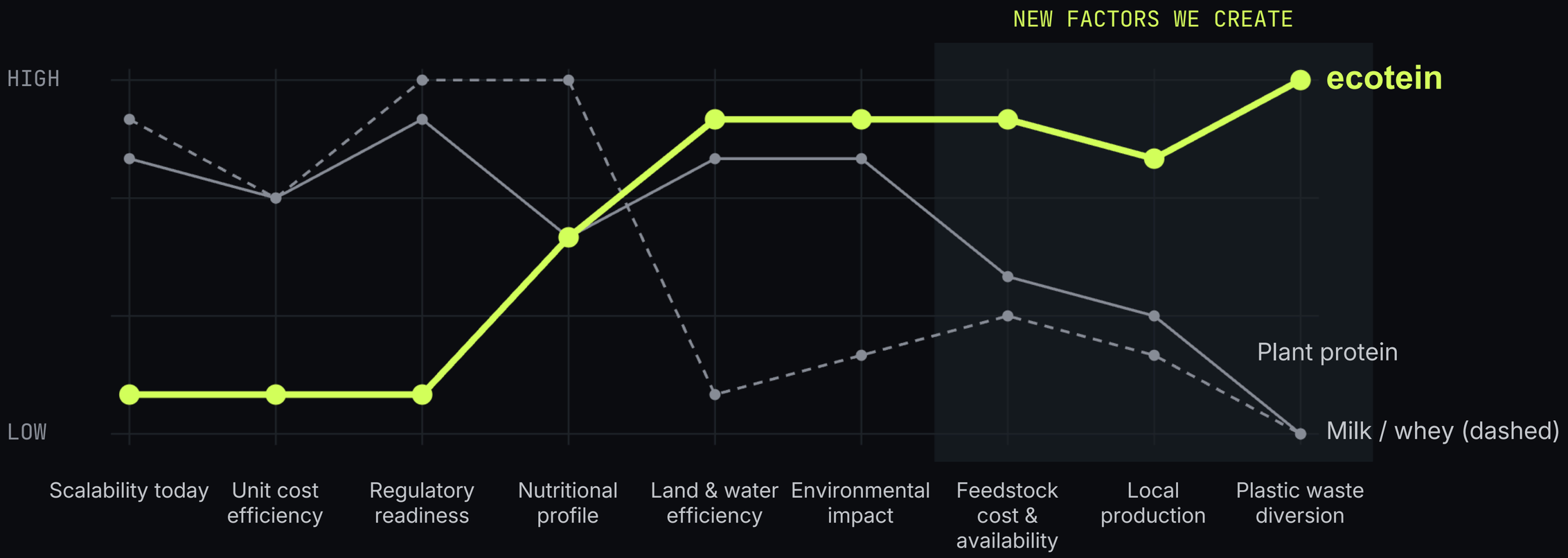
SOM $15 M

Fish feed only: Priced 8% below fishmeal per unit of protein.

Sources: Grand View Research · GMI · WFP · UNICEF · Lifemate

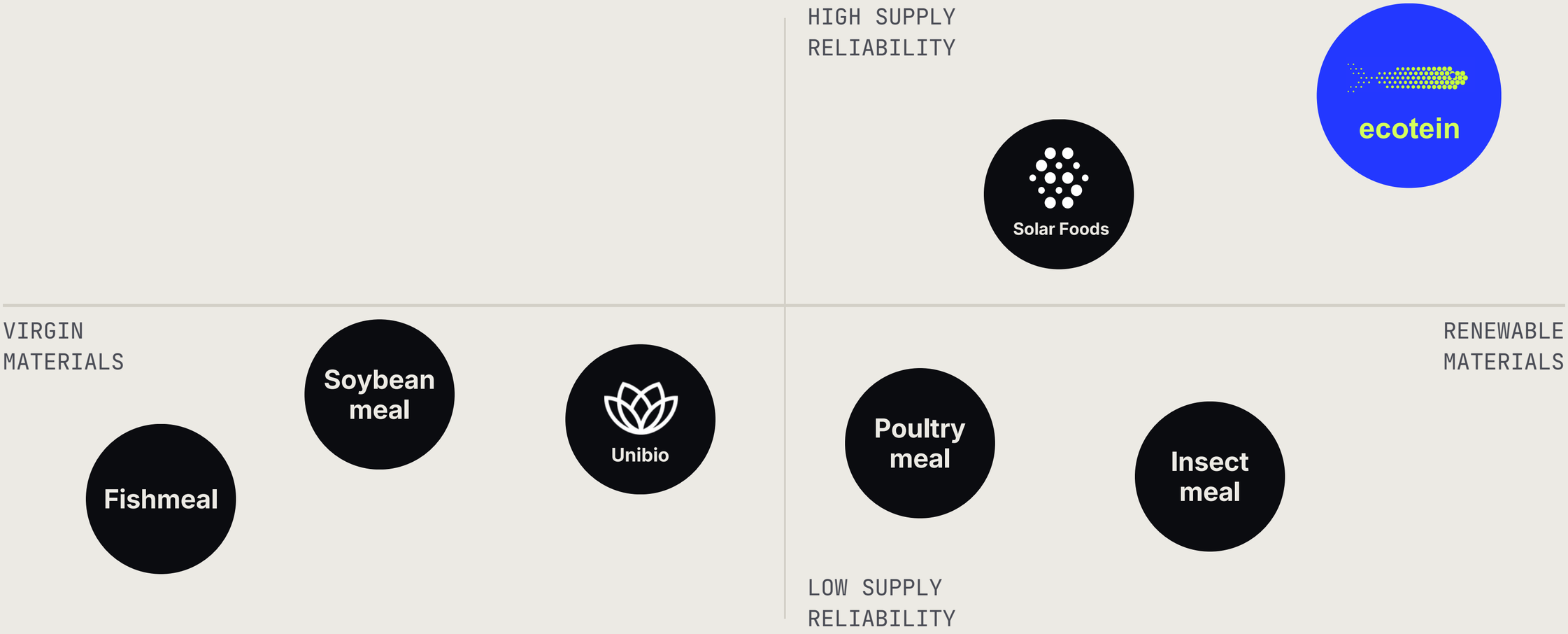
Blue Ocean Strategy

Ecotein is an innovative product that creates a new market altogether.



Competitor Analysis

Renewable, resilient, and **disruptive**.



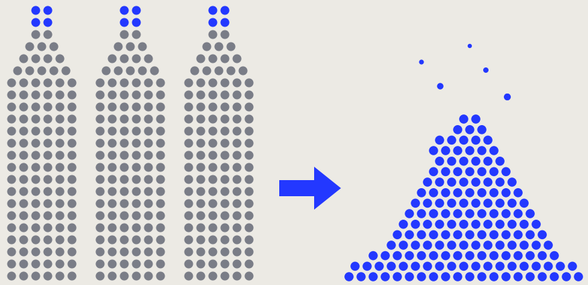
Protein Unlike Any Other

No Farmland, no wild fish. Just plastic.

~7,865 t

PROTEIN

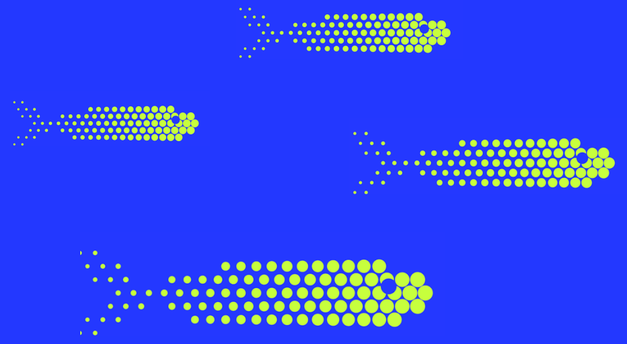
a year, from 14,300 t of plastic



~25,000 t

FOOD-GRADE WILD FISH

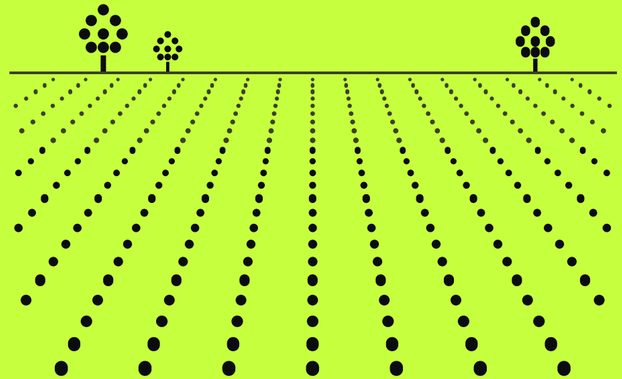
left in the sea a year, not ground into fishmeal



~10,600

ACRES OF FARMLAND NOT NEEDED

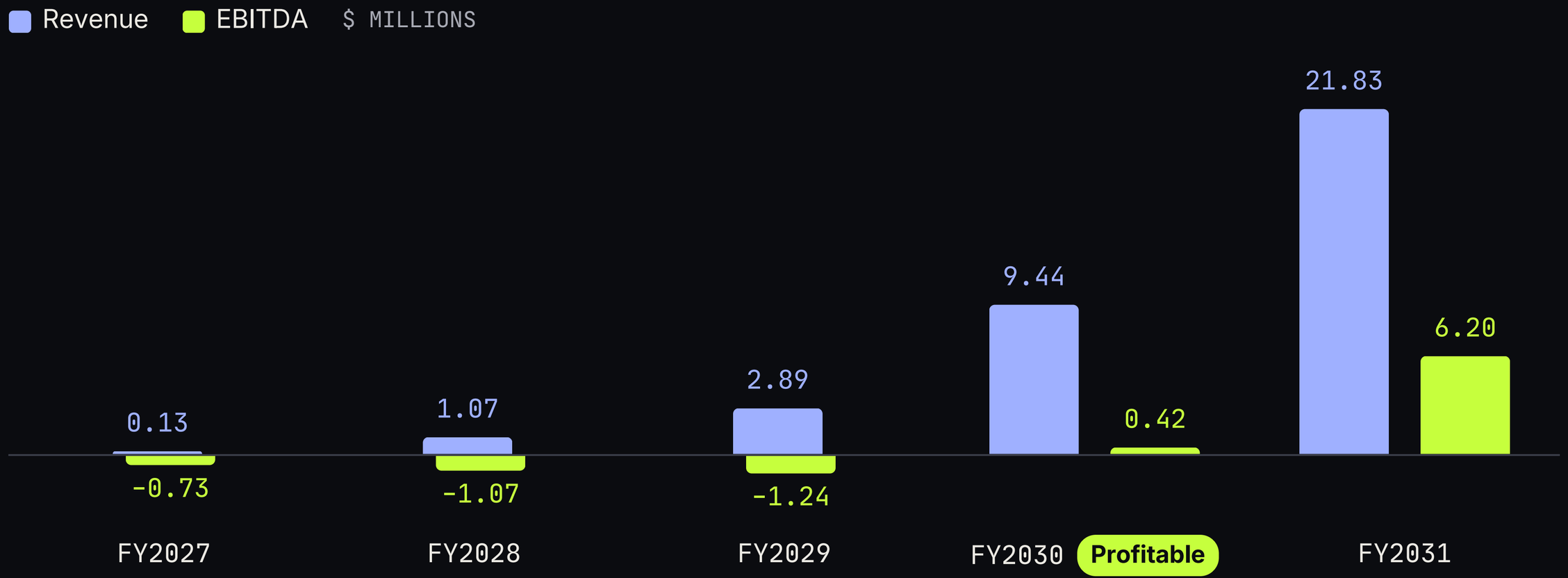
a year, about 6% of Singapore's land



Figures per year, 2031

Financial Overview

Profitable in 2030, $6.2M EBITDA by 2031



Gross margin 50% in 2031. Lowest cash point −$42.6M in 2029. Assumes $2,300/t, 55% yield, free plastic.

Financial Use of Funds

We're raising $50M in two stages, each unlocked by proof.

$14M STAGE 1 · 2027–2028

Plant 1 and first sales. Proves:

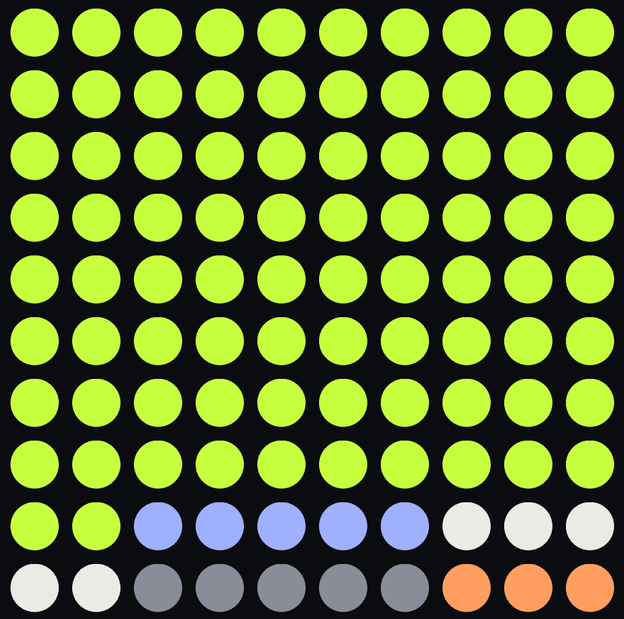
- Protein spec and a pilot yield of 35%+
- Paid feed trials with mills

$36M STAGE 2 · 2028–2030

Plant 2 and profitability. Proves:

- Supply contracts with feed mills
- Positive EBITDA in 2030

STAGE 1 USE OF FUNDS · EACH DOT = 1%



Plant 1 build	$10.50M
Team	$0.64M
Regulatory & R&D	$0.59M
Plant costs & pilots	$0.58M
Marketing & legal	$0.41M

Full pro forma: Pro Forma Files on Google Drive

Product Team 10

Five builders across engineering, design, and business



Yasmin Aref

SOFTWARE ENGINEERING



Paul Kang

SOFTWARE ENGINEERING



Seraph Kumar

PRODUCT DESIGN



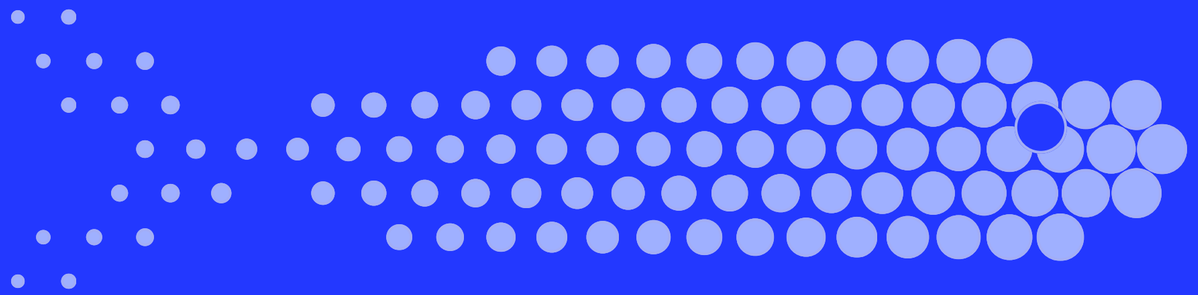
Adara Lui

BUSINESS & DESIGN



Naomi Meininger

SOFTWARE ENGINEERING



Thank you.

Plastic In → Protein Out

Yasmin Aref · Paul Kang · Seraph Kumar · Adara Lui · Naomi Meininger

Competitor Comparison Chart

Where Ecotein stands today

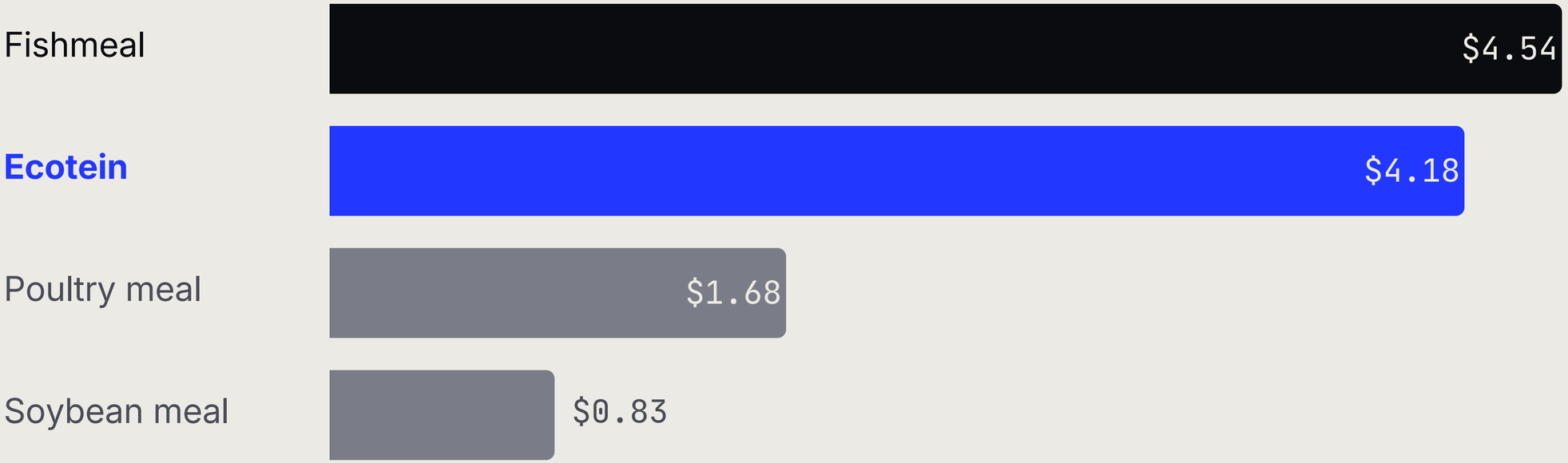
	No farm or fish	Paid for input	Cleared for sale	Scale today
Ecotein	✓	✓*	× (not yet)	× (not yet)
Solar Foods	✓	×	✓ (SG)	✓
Unibio	✓	×	✓ (KSA)	✓
Insect meal	Partly	Partly	✓	Partly
Poultry meal	×	×	✓	✓
Soybean meal	×	×	✓	✓
Fishmeal	×	×	✓	✓

*A hypothesis: gate fees are off in our base case. SG = Singapore, KSA = Saudi Arabia (fish feed).

Price Benchmark

We compete with fishmeal, not soy

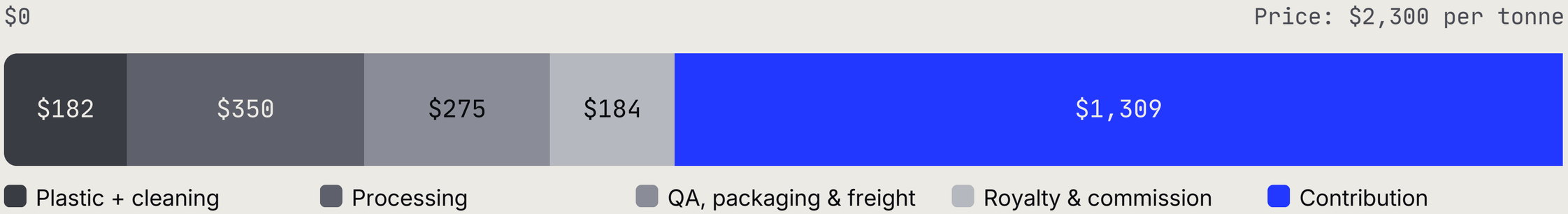
COST PER KG OF PROTEIN, US$



8% cheaper than fishmeal per kg of protein, the price fish-feed mills buy against.

Unit Economics

Every tonne of fish feed earns $1,309



57%
contribution margin on fish feed in 2031 (34% in 2027)

$0
plastic purchase cost: we take low-value PET recyclers reject (was $450/t)

$1.4M
2031 cost of every extra $100/t spent cleaning plastic

Price Sensitivity

What if fishmeal falls or yield misses?

CASE	ASSUMPTIONS	PRICE	EARNED, 55% YIELD	EARNED, 45% YIELD
Low	Fishmeal back at early-2025 levels, 55% protein, 15% discount	$1,150	−$208	−$315
Base	Today’s fishmeal, 55% protein, about 8% discount	$2,300	$850	$743
High	Today’s fishmeal, 65% protein measured, 10% discount	$2,650	$1,172	$1,065

Earned = per tonne of fish feed after plant costs, royalty and commission, including plastic credits.

$1,770/t
break-even fishmeal, 55% yield

$1,900/t
break-even fishmeal, 45% yield

$2,951/t
fishmeal today (Sep 2026)

Why not poultry or pig feed? Soy-priced protein is worth ~$460/t to a mill, below our ~$800/t cost.

Financial Use of Funds

Where the $50M goes

STAGE 1 USES, JAN 2027–AUG 2028	AMOUNT	SHARE
Plant 1 equipment and build	$10.50M	82.5%
Plant staff, rent, maintenance and pilot lots	$0.58M	4.6%
Founders, core team and added hires	$0.64M	5.0%
Regulatory, SFA dossier, trials, R&D and license	$0.59M	4.6%
Marketing, legal, insurance, pilot rent, commissions	$0.41M	3.2%
Gross uses	$12.72M	100%
Less revenue earned and grants	−$1.39M	
Plus 15% contingency and working capital	$1.70M	
Stage 1 total (raise rounded up to $14M)	$13.03M	

STAGE 2 · SEP 2028 →

$36M

Series B or project debt backed by offtake agreements. Pays for Plant 2 ($30M, 10,000 t/yr, Sep 2028–Jun 2029) and the run to profitability in 2030.

Market Math

How we sized the market

ITEM	VALUE	SOURCE OR METHOD
Global fish feed and pet food, 2025	55.5M t and 39.3M t	Alltech Agri-Food Outlook 2026
TAM: all their protein, as our powder	40.3M t × $2,300 + 12.9M t × $4,250 = $147B	40% protein in fish feed; 18% in pet food (AAFCO minimum); 55% in our powder
Global fishmeal supply, 2026	under 4.6M t	IFFO forecast (The Fish Site, Oct 2026)
SAM: 30% of Asia-Pacific fishmeal	0.82M t of powder = $1.9B	Asia-Pacific ≥50% of use (IFFO) × 30% replaceable (team assumption)
SOM: 2031 fish-feed sales	6,546 t = $15.1M	0.8% of SAM
Singapore demand, high case	about 740 t a year	6,800 t local fish × 1.5 feed ratio × 40% protein × 10% replaced
Pet food (incl. ornamental), 2031	$5.6M	Pro forma; ornamental fish feed is a ~$323M global market (2025)

Expert Interview Insights

What experts told us, and what we changed

BUSINESS CONSULTANT · OPERATIONS, FINANCE

“The disruptive angle could make this more financially viable than existing ways of reducing plastic.”

WHAT WE CHANGED

We lead with the business case: free plastic feedstock, and plastic credits as a second income.

SOFTWARE ENGINEER · INDUSTRIAL, DATA VISUALIZATION

“Misinformation could scare buyers away, but strong branding and education could help us ride the protein wave.”

WHAT WE CHANGED

We sell business to business through feed mills, with safety data first, not a consumer brand.

PROCESS DEVELOPMENT ENGINEER · BIOTECH MANUFACTURING

“Scaling is the hard part: commercial production needs bioreactors of roughly 10,000–100,000 L.”

WHAT WE CHANGED

We start with a contract fermenter and build our own plants only from 2028.

HEALTHCARE PROFESSIONAL · PUBLIC HEALTH

“Safety data will decide adoption: what happens to the plastic’s contaminants and additives.”

WHAT WE CHANGED

A feedstock safety package is a product requirement, and it comes before any sale.

Sources

Fishmeal price, Apr–Sep 2026: YCharts, Peru fish meal price (World Bank / IMF)

2026 fishmeal supply: IFFO via The Fish Site, 8 Oct 2026

China's share of fishmeal use: IFFO, "Feeding a Dragon"

Plastic recycling: 9% global, 5% Singapore: OECD (2022); NEA via Eco-Business (2024)

Incineration fees: NEA, refuse disposal fees (2026)

Feed rules; pet food scope: Singapore Food Agency, animal feed page

Bottle deposit scheme: Packaging Gateway, 1 Apr 2026

Plastic credit prices: World Bank PROBLUE (2024); Eunomia (2024)

Ornamental fish feed market: LP Information, 2025–2031 forecast

Singapore ornamental fish exports: Mothership, Dec 2017

BioPROTEIN research: Michigan Tech News; MTU 2022 PET review; EMD Future Insight Prize 2021

Competitors: Singapore EDB; IntraFish; SFA Approved Novel Foods (Mar 2026)

Grants and contract fermentation: Enterprise SG and NUS Bezos Centre; A*STAR; Singapore EDB

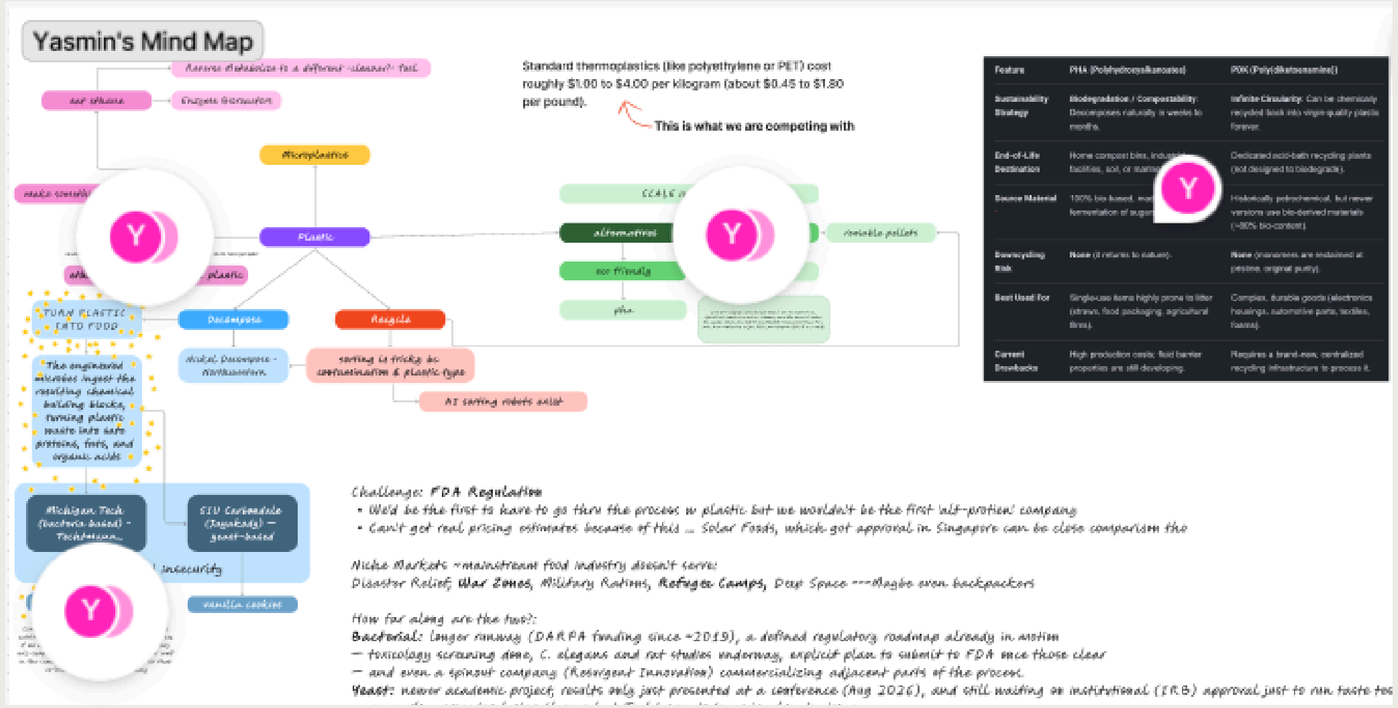
Poultry and soybean meal prices: Fastmarkets (Sep 2026); World Bank via Protein Report (Jul 2026)

Krill meal prices: aquafeed.com

All financials: Ecotein_Pro_Forma.xlsx (team model, revised 31 Oct 2026)

Product Team Process

Research, Define, Ideate, Iterate



Yasmin's Mind Map; Link to all team members': figma.com/board/9s9oKDsggISmAa9tef0xR7

01 Surface
Each member researched potential challenges of interest to share with the team.

02 Align
The team aligned on one challenge area before ideating on potential solutions.

03 Diverge
Each member built a mind map around potential drivers, resources, and constraints.

04 Align
The team shared findings before settling on the problem space and framing.

05 Ideate & Iterate
The team continually collaborated to develop a research-backed solution for the problem.

AI Usage

Our product team used Claude (Anthropic) to restyle this deck to our style guide, conduct market research (fishmeal prices, Singapore rules, plastic credits, market sizes), and refine our pro forma to focus on fish-feed and pet-food.

Our team set the direction, reviewed every output and checked figures against the sources listed. Assumptions the AI introduced are labeled in the pro forma for us to verify.

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